A Causal Measurement of Anti-Fingerprinting Defense Effectiveness in the Real-World Ad Ecosystem

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Abstract

Third-party cookie deprecation and client-side storage partitioning have accelerated the commercial deployment of stateless browser fingerprinting across programmatic advertising ecosystems. Existing empirical benchmarks evaluate anti-fingerprinting defenses through passive script detection or synthetic Shannon entropy calculations, metrics that capture tracking intent or theoretical variance, but fail to establish whether defenses prevent client re-identification in production ad tech.

We present an outcome-oriented causal measurement framework that uses programmatic ad-auction clearing prices as an empirical ground-truth oracle for identity certainty, conducting the first systematic evaluation of modern anti-fingerprinting defenses in production ad tech. Deploying an automated infrastructure across 150 commercial publishers under native browser execution, we enforce a three-phase intervention: training high-intent behavioral profiles, capturing baseline Prebid.js bids, purging all client-side stateful storage, and measuring post-purge bid trajectories across five hardened and baseline browser configurations.

Our framework reveals that stateful clearing alone is completely ineffective: unhardened Google Chrome exhibits an unperturbed median bid shift of -1.12% ($p = 0.4832$, Re-ID rate $R_{\text{ID}} = 0.606$), proving commercial Demand-Side Platforms (DSPs) seamlessly bridge storage resets via stateless fingerprints. Conversely, anti-fingerprinting defenses construct an empirical Pareto frontier. Hardened Mozilla Firefox (combining RFP normalization with state partitioning) emerges as a zero-cost defense, yielding a -26.87% median bid collapse ($p = 0.0734$) with zero runtime exception overhead ($U_{\text{cost}} = 0.00$). Hardened Brave Browser achieves the apex of tracking disruption (-59.97% median collapse, $R_{\text{ID}} = 0.500$), but imposes a $+44\%$ increase in runtime script breakage ($U_{\text{cost}} = 0.44$).

Finally, we deconstruct the "Usability Trap", showing how defensive network blocking triggers tracker retry loops that distort naive tracking metrics and formulate the Refined Effectiveness-to-Risk metric (ETR*), providing a mathematically robust benchmark for privacy-preserving web engineering.

CCS Concepts

• **Security and privacy** → **Usability in security and privacy**; **Web protocol security**; • **Information systems** → *Web applications.*

Keywords

Browser Fingerprinting, Web Privacy, Real-Time Bidding, Causal Inference, Programmatic Advertising, Usability

1 Introduction

For two decades, the commercial web's surveillance economy operated on stateful storage primitives: HTTP cookies, Flash Local Shared Objects, and HTML5 Web Storage [7, 8]. In response to consumer pushback and regulatory mandates, major browser vendors introduced storage partitioning architectures, notably Apple WebKit's Intelligent Tracking Prevention (ITP) and Mozilla Gecko's Total Cookie Protection (dFPI). As third-party cookies decay, tracking consortiums have systematically pivoted to **stateless browser fingerprinting**: interrogating hardware, driver, and OS-level attributes through Web APIs (Canvas, WebGL, AudioContext, WebRTC, and system font enumeration) to reconstruct persistent cross-site behavioral graphs [9, 11].

To counteract stateless tracking, modern browsers deploy two fundamentally diverging anti-fingerprinting philosophies:

(1) **API Normalization:** Emulated by Tor Browser and Mozilla's Resist Fingerprinting (`privacy.resistFingerprinting`, RFP), which quantizes high-resolution timers, spoofs system attributes to standardized pools, and forces canvas readbacks to uniform white screens to maximize client anonymity sets [6].
(2) **API Randomization (Farbling):** Implemented by Brave Browser, which injects cryptographically pseudo-random, session-ephemeral noise into canvas rendering, audio buffers, and hardware device enumeration, ensuring that subsequent readbacks generate divergent hashes that disrupt cross-session linking [12].

Despite these deployments, the measurement literature lacks a rigorous methodology to determine whether these defenses functionally protect users in production. Current empirical benchmarks suffer from two structural flaws:

The Intent vs. Outcome Fallacy. Passive measurement platforms (e.g., OpenWPM [8], FP-Inspector [9]) instrument JavaScript APIs to log when heuristic execution patterns occur. While effective for detecting tracking *intent*, observing a canvas readback provides zero verification of tracking *success*. It cannot determine whether the extracted payload resolved a user within an ad-tech identity graph or failed to overcome environmental noise.

The Theoretical Entropy Gap. Synthetic benchmarks (e.g., Panopticlick [7], AmIUnique [11]) benchmark defenses by calculating the Shannon entropy bits subtracted from isolated Web APIs. However, theoretical entropy reductions do not guarantee tracking severance in the wild. If an advertising network combines a partially farbled canvas signature with residual cross-site signals, such as IP-range localization, TLS client fingerprints, and User-Agent headers. Client re-identification persists despite nominal entropy loss.

Real-Time Bidding as an Economic Oracle. To overcome the limitations of proxy measurements, we exploit a direct side-channel oracle: **programmatic Real-Time Bidding (RTB) clearing prices** [12]. In header bidding auctions orchestrated via client-side Prebid.js wrappers, algorithmic Demand-Side Platforms (DSPs) submit bids (Cost Per Mille, or CPM) within milliseconds based on identity certainty [13]. Econometric studies establish that an identified user commanding a rich cross-site behavioral profile sells for a 30% to

60%+ valuation premium over an untargeted contextual impression [3, 10, 15].

When an anti-fingerprinting defense successfully breaks cross-site tracking across an enforced stateful reset, DSP identity graphs are severed. Lacking identity certainty, bidding algorithms retreat to contextual reserve floors, precipitating a measurable collapse in CPM clearing prices.

Contributions. To quantify this phenomenon, we built a cluster-scale causal measurement platform that drives native Linux browser binaries across 150 commercial publishers. Our work delivers four primary contributions:

- **Causal Measurement Methodology:** We design a 3-phase experimental protocol that trains high-intent personas, records pre-break ad-auction baselines, programmatically purges stateful storage, and measures post-purge economic bid shifts, causally isolating stateless tracking in the wild.
- **Validation of the Negative Control:** We demonstrate that in unhardened Google Chrome, stateful storage purges yield a negligible bid shift of only -1.12% ($p = 0.4832$, fail to reject H_0). This provides empirical proof that clearing cookies without anti-fingerprinting defenses is entirely futile against commercial ad tech.
- **The Empirical Pareto Frontier:** We characterize an operational trade-off in browser engineering: *Hardened Firefox* achieves the **Stealth / Zero-Breakage Optimum** (-26.87% median bid collapse, $p = 0.0734$, with zero usability breakage: $U_{\text{cost}} = 0.00$); whereas *Hardened Brave* reaches the **Maximum Privacy Optimum** (-59.97% median collapse, $R_{\text{ID}} = 0.500$), but imposes a $+44\%$ increase in runtime script exceptions ($U_{\text{cost}} = 0.44$).
- **Deconstructing the Usability Trap & Metric Refinement:** We show that defensive request blocking triggers tracker retry loops (28,976 requests in Firefox Hardened) that invert naive evaluation metrics. We formalize the Refined Effectiveness-to-Risk metric (ETR*), resolving structural pathologies and restoring monotonic rigor to privacy defense evaluation.

2 Background & Threat Model

2.1 Programmatic Advertising & Header Bidding

Modern web publishers monetize digital inventory via programmatic Real-Time Bidding (RTB). While legacy display monetization relied on monolithic ad servers (e.g., Google Ad Manager), modern publishers deploy **Header Bidding** wrappers, principally the open-source `Prebid.js` framework, embedded directly in the client-side DOM [5, 13].

As illustrated in Figure 1, when a client navigates to a publisher:

(1) `Prebid.js` dynamically reads configuration tags, identifies eligible ad slots, and issues parallel, asynchronous HTTP bid requests to configured Supply-Side Platform (SSP) adapters (e.g., Rubicon, OpenX, AppNexus).
(2) Each SSP adapter forwards the bid request via OpenRTB protocols to participating Demand-Side Platforms (DSPs). This request encapsulates client attributes: geographical metadata, viewport dimensions, cookie-sync identifiers, and stateless device fingerprint signals collected by embedded tracker scripts [14].

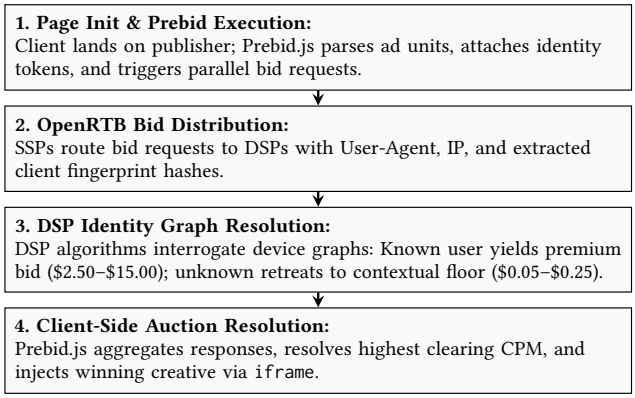


Figure 1: Client-side Prebid.js auction lifecycle.

(3) DSP bidding algorithms parse the payload and query internal identity graphs. If the payload matches a high-intent user persona, the DSP bids aggressively. If the payload cannot be linked to historical data, the DSP bids at contextual reserve floors [2].
(4) Winning bids return to the browser, where `Prebid.js` resolves the highest clearing CPM, surfaces it to the publisher's ad server, and renders the ad creative.

2.2 The Economic Targeting Premium

The bedrock econometric assumption of digital ad tech is that identified users command significantly higher prices than anonymous users. Longitudinal empirical studies by Beales and Eisenach demonstrated that behavioral-targeted advertising sells for a 2.68× price premium over non-targeted advertising (+168%) [3].

Subsequent field trials by Google confirmed that stripping user identifiers precipitously decreases publisher ad revenue by an average of 52% [15], a finding echoed by Johnson et al. (54% revenue decline) [10].

Consequently, we formalize clearing CPM as an *economic side-channel oracle*:

$$\text{CPM}(u) = \alpha \cdot \mathbb{I}_{\text{identified}}(u) + \beta_{\text{context}} + \epsilon_{\text{market}} \tag{1}$$

where α represents the behavioral targeting premium ($\alpha > 0$), β_{context} represents base contextual inventory value, and ϵ_{market} denotes stochastic auction noise.

2.3 Threat Model

Adversarial Capabilities. We model commercial tracking consortiums, SSPs, and DSPs as curious-but-honest commercial observers. The adversary operates JavaScript code embedded within publisher domains, participates in programmatic auctions, and seeks to persist user identities across disjoint first-party contexts ($s_i \neq s_j$).

The adversary has unrestricted access to standard browser Web APIs and executes stateless fingerprinting techniques (Canvas pixel extraction, WebGL parameter enumeration, AudioContext oscillator decay, system font profiling, and WebRTC candidate enumeration).

Adversarial Boundaries. The adversary *cannot* compromise the underlying operating system, inspect kernel or process memory, forge TLS cryptographic primitives, or mount machine-in-the-middle network tampering. IP-based location matching is constrained to geographic subnet clustering.

Client Defenses. The client executes a modern web browser equipped with anti-fingerprinting defenses (API normalization, API farbling, and defensive network filtering). The user's objective is to enforce an *identity break* by programmatically flushing stateful storage and verifying that stateless defenses prevent re-identification.

3 Measurement Platform Architecture

Automated privacy measurement platforms frequently succumb to the **Emulation Fallacy** [4]: headless automation frameworks (e.g., Puppeteer, vanilla Selenium) leak automated execution signatures (`navigator.webdriver`, missing rendering contexts, anomalous clock skews).

Commercial ad-fraud filters detect these anomalies and drop bid requests, causing crawlers to record artificial auction artifacts [1]. To guarantee experimental fidelity, we designed and deployed a hardened, distributed measurement cluster running native Linux binaries over virtual displays.

3.1 Cluster Execution & Environment Hardening

Batch Orchestration. The harness is deployed across dedicated HPC cluster nodes via the OAR batch scheduler. Worker nodes execute isolated Docker containers implementing a *sidecar proxy architecture*.

Native Binaries on Virtual Framebuffers. Instead of relying on Chromium's `–headless=new` or Gecko's `-headless` flags, both of which bypass standard graphics pipelines and alert anti-bot heuristics [4], the platform executes uninstrumented native Linux binaries (Google Chrome, Mozilla Firefox, Brave Browser). Browsers run within an isolated X Virtual Framebuffer (`Xvfb`) display server operating at display `:99` ($1920 \times 1080 \times 24$, 24-bit TrueColor depth).

Mesa Software Rasterization. Cluster worker nodes feature heterogeneous physical GPUs, which would introduce divergent hardware entropy into WebGL and 2D canvas draw operations. We eliminate physical GPU confounding by forcing Mesa software rasterization across all containers:

```
export LIBGL_ALWAYS_SOFTWARE=1
export MESA_LOADER_DRIVER_OVERRIDE=swrast
```

This forces Chromium and Gecko rendering engines to execute all graphics operations via identical CPU-based rasterization logic (`swrast/llvmpipe`), ensuring baseline visual determinism across cluster workers.

D-Bus Nullification. Containerized browser processes frequently hang when background threads attempt to communicate with desktop secrets daemons (GNOME Keyring, KWallet) over system D-Bus channels. We remediate this by unsetting `DBUS_SESSION_BUS_ADDRESS`, binding socket addresses to `/dev/null`, and generating unique, container-isolated machine identifiers via `dbus-uuidgen` during entrypoint initialization.

3.2 Network Interception & Proxy Stealth

Network interception and auction event harvesting are managed via an upstream `mitmproxy` sidecar instance incorporating specific stealth countermeasures:

Dual-Tier CA Injection. To inspect TLS flows without launching browsers under `--ignore-certificate-errors` (a flag that strips browser security guarantees and alerts commercial crawlers), our platform installs an ephemeral root CA into both:

(1) The host operating system trust store (`/usr/local/share/ca-certificates/` via `update-ca-certificates`);
(2) The user NSS SQLite databases (`/root/.pki/nssdb` via `certutil -A -t"TC,,"`).

Transport-Layer Fingerprint Evasion. Python-based proxy engines (such as `mitmproxy`'s HTTP/2 implementation) emit distinctive frame multiplexing and window-update sequences that commercial Web Application Firewalls (Cloudflare, Akamai) classify as crawler traffic. We pass `–no-http2` to the proxy, downgrading upstream network transport to clean HTTP/1.1 pipelines and neutralizing proxy-specific TLS/H2 fingerprinting.

Causal Anti-Cache Enforcement. To guarantee that Phase 3 measurements represent genuine live network transactions rather than disk artifacts, the proxy strips `If-Modified-Since` and `If-None-Match` headers via `–anticache`. This forces origin servers to return explicit HTTP 200 responses, preventing HTTP 304 Not Modified responses from fabricating tracking persistence.

Out-of-Band IPC & Referrer Concealment. The automation harness communicates with the interception proxy using the unallocated non-routable IP address `240.240.240.240`. To navigate between publisher targets without leaking control metadata into browser history or polluting outbound `Referer` headers, the proxy injects an inline HTML anchor:

```
<a href="{target_url}" rel="noreferrer"
   id="autoclick"></a>
```

A 100ms timer triggers this synthetic click, concealing the control endpoint. Unhandled runtime script exceptions intercepted by driver listeners are transmitted out-of-band via Python's native `urllib.request` directly to `http://240.240.240.240/js_error`.

3.3 3-Tier Hybrid CMP Auto-Consent Engine

Under European GDPR and ePrivacy frameworks, programmatic header bidding auctions stall until user consent is recorded. Unhandled Consent Management Platform (CMP) dialogs freeze `Prebid.js` bid loops, resulting in complete auction drops. To resolve this, we engineered an asynchronous 3-tier hybrid consent engine: (1) **Tier 1 (API-First Settlement):** Inspects the global `window` context and directly triggers official framework handlers (`OneTrust.AllowAll()`, `Didomi.setUserAgreeToAll()`, or `Cookiebot.dialog.submit()`); (2) **Tier 2 (Targeted CSS Mutators):** Dispatches synthetic mouse events targeting standard European consent selectors (`#onetrust-accept-btn-handler`, `.cmp-agree-all`, or `button[id*="consent"]`) combined with programmatic wheel scrolling; (3) **Tier 3 (Heuristic DOM Regex Traversal):** Recursively traverses the DOM and consent `iframe` scopes to match affirmative multilingual strings (English, Portuguese, Spanish: e.g., `/(AcceptAll|AceitarTodos|AceptarTodo)/i`) bounded strictly by character length ($2 \leq \text{length} < 35$).

3.4 Atomic Profile Sanitization

In preliminary trials, standard profile cloning leaked persistent identifiers across experimental phases. Specifically, Blink-based browsers cache session descriptors and identity graphs within LevelDB binary stores, HSTS caches, and persistent Service Worker storage, producing an artificial 0.0% bid delta.

We resolved this by developing an **Atomic Profile Sanitizer**. To construct the clean profile for Phase 3, the engine extracts configuration files while stripping stateful directories using strict exclusion patterns:

```
EXCLUDED_PATTERNS = shutil.ignore_patterns(
    'SingletonLock', 'Cookies*', 'Cache*',
    'Local Storage', 'IndexedDB', 'Sessions',
    'Session Storage', 'TransportSecurity',
    'Network Persistent State', 'QuotaManager',
    'Service Worker', 'shared_proto_db'
)
```

This guarantees that the post-break browser initializes from a provably clean storage slate while preserving exact defense configurations.

4 Causal Measurement Methodology

To evaluate anti-fingerprinting defenses without relying on static code heuristics or theoretical entropy proxies, we formulate our measurement as an empirical causal inference pipeline. By training high-value behavioral personas and capturing clearing prices in programmatic ad auctions, we test whether clearing stateful storage causes bid values to drop to generic floor levels (indicating tracking disruption) or remain elevated (proving stateless identity persistence).

4.1 Causal Experimental Design

Let $\mathcal{S}$ denote the set of evaluated commercial publisher domains ($|\mathcal{S}| = 150$), and let C represent the set of evaluated browser configurations. For each configuration $c \in C$, our cluster infrastructure executes a deterministic 3-phase protocol:

Phase 1A: Persona Training ($t_0 \rightarrow t_1$). A fresh browser profile navigates across 14 high-value commercial seed domains structured across four advertising verticals: (1) *Financial & Insurance (High CPM):* geico.com, progressive.com, statefarm.com, allstate.com, nerdwallet.com, investopedia.com; (2) *Automotive Research (High Intent):* kbb.com, autotrader.com; (3) *Luxury Retail & Lifestyle (Retargeting):* architecturaldigest.com, robbreport.com; (4) *High-End Travel & Real Estate:* fourseasons.com, ritzcarlton.com, remax.com, century21.com. On each domain, the crawler resolves consent via the 3-Tier Consent Engine, dispatches hardware-level cursor movements, and waits for tracking pixels to register, establishing an "Affluent Consumer" profile in commercial device graphs.

Phase 1B: Pre-Break Baseline Measurement ($t_1 \rightarrow t_2$). Holding this persona, the browser navigates the 150 publisher targets. Upon reaching each domain, the crawler dispatches synthetic cursor events, resolves consent, and maintains a strict **15.0-second auction observation window** to allow asynchronous Prebid bid adjustments to settle. The network proxy intercepts clearing prices:

$$\mathrm{CPM}_{\mathrm{Pre}}(s) = \max_{b \in \mathcal{B}_s} \{\mathrm{CPM}(b)\} \tag{2}$$

where $\mathcal{B}_s$ is the set of all valid bids returned across active SSP adapters.

Phase 2: The Enforced Identity Break ($t_2 \rightarrow t_3$). The browser process is closed, releasing operating system file locks. The Atomic Profile Sanitizer purges all stateful directories (`Cookies*`, `Local Storage`, `IndexedDB`, `Session Storage`, `Sessions`, `Cache*`, and NSS databases), while preserving extension configurations, anti-fingerprinting shields, and browser preferences.

Phase 3: Efficacy Measurement ($t_3 \rightarrow t_4$). The sanitized browser revisits the exact same 150 publisher domains in identical order under the same network routing. The platform records the post-break clearing price:

$$\mathrm{CPM}_{\mathrm{Post}}(s) = \max_{b \in \mathcal{B}'_s} \{\mathrm{CPM}(b)\} \tag{3}$$

alongside runtime JavaScript exceptions and network tracking flows.

4.2 Publisher Curation Pipeline

To ensure auctions are observable, publishers must run client-side header bidding rather than opaque server-side auctions [5, 13]. We curated $\mathcal{S}$ ($|\mathcal{S}| = 150$) from top Tranco ledger entries via a 4-stage filter: (1) **TLD Selection:** Restrict domains to primary commercial English TLDs (`.com`, `.net`, `.org`, `.uk`, `.ca`, `.au`, `.nz`, `.ie`); (2) **Walled-Garden Elimination:** Filter out a 40-domain blacklist of platforms that monetize inventory exclusively through proprietary internal ad tech without OpenRTB (e.g., Google, Facebook, Amazon, Apple, Netflix, Wikipedia, StackOverflow); (3) **Header Bidding Signature Detection:** Inspect landing-page source HTML for active wrappers (`prebid.js`, `pbjs.que`, `pubfood`, `apstag.js`, or `gpt.js`); (4) **Supply-Chain Validation (`ads.txt`):** Query `https://{domain}/ads.txt` and require an HTTP 200 status containing `"DIRECT"` authorized exchange relationships.

The Paired Observability Filter ($\mathcal{S}_{active}$). Because production auctions can suffer transient exchange timeouts or dynamic allocation shifts, we enforce a strict Paired Observability Filter:

$$\mathcal{S}_{\mathrm{active}}(c) = \{s \in \mathcal{S} \mid \mathrm{CPM}_{\mathrm{Pre}}(s) > 0 \ \wedge \ \mathrm{CPM}_{\mathrm{Post}}(s) > 0\} \tag{4}$$

Across configurations, $|\mathcal{S}_{\mathrm{active}}|$ ranges from 33 to 54 domains (22.0% to 36.0% transparency), ensuring that all calculated deltas represent valid, paired transactions. Workloads were parallelized across three Docker containers (50 domains/batch, 4 GB shared memory) on dual Intel Xeon E5-2609 v4 and AMD EPYC 7343 nodes.

4.3 Evaluated Browser Configurations

We evaluate five browser configurations across Blink and Gecko rendering engines: (1) **`chrome_baseline`:** Vanilla Google Chrome operating under default settings, acting as our unhardened control group; (2) **`firefox_baseline`:** Standard Mozilla Firefox with Enhanced Tracking Protection (ETP Standard); (3) **`firefox_hardened`:** Multi-layered hardened Firefox combining ETP Strict (`browser.contentblocking.category="strict"`), Total Cookie Protection (`network.cookie.cookieBehavior=5`, dFPI), network state partitioning, query parameter stripping, and Tor-derived Resist Fingerprinting (`privacy.resistFingerprinting=true`); (4) **`brave_baseline`:** Standard Brave Browser with default Shields (Canvas and Audio

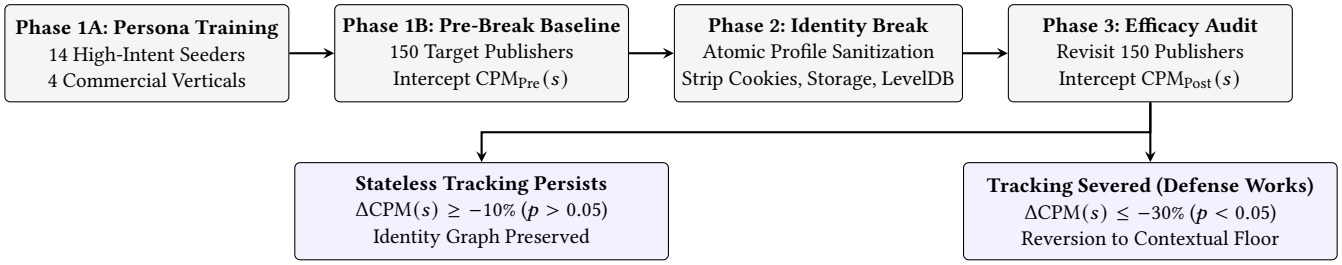


Figure 2: The 3-Phase Causal Measurement Lifecycle: Persona training ($t_0 \rightarrow t_1$), Pre-Break baseline capture ($t_1 \rightarrow t_2$), programmatic profile sanitization ($t_2 \rightarrow t_3$), and Post-Break re-crawl ($t_3 \rightarrow t_4$).

Farbling enabled; standard network blocking); (5) **`brave_hardened`:** Brave Browser with "Strict" Shields (farbling combined with aggressive network request blocking).

4.4 Metrics & Econometric Ground Truth

Relative Economic Bid Shift (ΔCPM). For each active domain $s \in \mathcal{S}_{\text{active}}$:

$$\Delta\text{CPM}(s) = \frac{\text{CPM}_{\text{Post}}(s) - \text{CPM}_{\text{Pre}}(s)}{\text{CPM}_{\text{Pre}}(s)} \tag{5}$$

Econometric Resistance Thresholds. Drawing on empirical ad-pricing literature, we partition observed responses into three regimes: (1) **Tracked / Low Resistance (ΔCPM $\geq -10\%$):** RTB auctions display 5%–10% natural volatility from DSP budget pacing and latency [2]; shifts within this boundary indicate stateless identity persistence. (2) **High Tracking Resistance (HTR, ΔCPM $\leq -30\%$):** Behavioral targeting yields a documented 30%–60%+ premium over untargeted contextual inventory [3, 10, 15]; drops exceeding 30% prove cross-site tracking was severed. (3) **Partial Resistance ($-30\% < \Delta$CPM $< -10\%$):** Represents intermediate valuation, partial graph matching, or algorithmic bid shading.

Re-Identification Rate (R_{ID}). The proportion of active publishers where the client is successfully re-identified:

$$R_{\text{ID}} = \frac{|\{s \in \mathcal{S}_{\text{active}} \mid \Delta\text{CPM}(s) \geq -10\%\}|}{|\mathcal{S}_{\text{active}}|} \tag{6}$$

Usability Cost (U_{cost}). Using the Site Breakage Index (SBI), runtime errors captured via our out-of-band logger are deduplicated by MD5 hashing normalized error strings and stack locations (U_e). To account for cross-engine variations between Gecko and Blink, we normalize against the control group (`chrome_baseline`):

$$U_{\text{cost}} = \max\left(0.0,\ \frac{U_e(\text{Test}) - U_e(\text{Chrome})}{U_e(\text{Chrome})}\right) \tag{7}$$

4.5 Statistical Hypothesis Testing

Because auction prices follow right-skewed, heavy-tailed distributions with left-censored reserve floors, parametric tests are invalid [2]. We apply the non-parametric **Paired Wilcoxon Signed-Rank Test**:

$$H_0 : \text{median}(\Delta\text{CPM}) = 0 \quad \text{vs.} \quad H_1 : \text{median}(\Delta\text{CPM}) < 0 \tag{8}$$

calculated over the signed ranks of non-zero absolute differences $|D_s| = |\text{CPM}_{\text{Post}}(s) - \text{CPM}_{\text{Pre}}(s)|$.

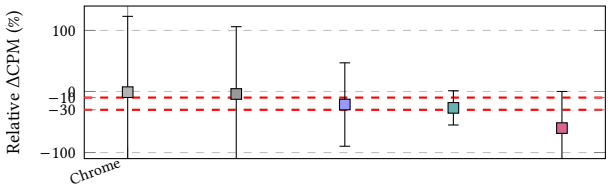


Figure 3: Distributional spread of relative price shifts (ΔCPM) across configurations. Points indicate median values; error bars denote the Interquartile Range (Q_1 to Q_3). Red dashed lines indicate the -10% noise and -30% HTR boundaries.

To test whether the five browser configurations diverge systematically across the complete parameter space, we compute an omnibus **Kruskal-Wallis test** across all groups.

5 Empirical Evaluation

Across all five browser configurations, our distributed measurement infrastructure achieved a **100.0% crawl yield**, evaluating all 150 target publishers across both experimental phases. Table 1 presents the empirical results across the three measurement silos: economic valuation distributions, network identity diffusion, and client-side usability overheads.

5.1 Dataset Integrity & CMP Consent Yield

Applying the Paired Observability Filter (Section 4.2) extracted between 33 and 54 actively observable publisher auctions per configuration (22.0% to 36.0% Prebid transparency). Crucially, the 3-Tier Consent Clicker successfully negotiated European privacy gates across the crawl:

- `chrome_baseline`: 107 domains resolved (71.3% auto-accepted; 61 regex, 43 CSS, 3 API);
- `firefox_hardened`: 118 domains resolved (78.7% auto-accepted; 64 regex, 53 CSS, 1 API);
- `brave_hardened`: 114 domains resolved (76.0% auto-accepted; 59 regex, 51 CSS, 4 API).

The unhandled minority (21.3%–32.0%) comprised non-EU sites lacking consent banners or publishers serving custom, non-standard modal overlays.

5.2 Silo 1: Economic Valuation Analysis

Figure 3 plots the full distributional spread of relative bid shifts across configurations, demonstrating how defensive hardening compresses market valuation.

Table 1: Master Empirical Evaluation Matrix across 150 Commercial Publishers. Absolute bids reflect Pre/Post medians; ΔCPM distributions show first quartile (Q_1), median, and third quartile (Q_3); Identity Diffusion decomposes into Cookie Syncs (S_{post}) and Smuggled UIDs (U_{post}); Usability Cost (U_{cost}) is normalized to the Chrome control group.

Configuration	$\lvert\mathcal{S}_{\text{active}}\rvert$	Pre $	Post $	R_{ID}	Spread (Q_1 / Med / Q_3)	IQR	S_{post}	U_{post}	D_{ID}	U_{cost}	Wilcoxon W (p)	ETR*
chrome_baseline	33	$0.08	$0.05	0.606	-75.00% / **-1.12%** / +123.11%	198.11%	3,206	18,359	21,565	0.00	226.5 (p = 0.4832)	0.105
firefox_baseline	46	$0.09	$0.05	0.543	-99.30% / **-4.11%** / +106.19%	205.49%	2,461	18,414	20,875	0.00	414.0 (p = 0.3445)	0.136
brave_baseline	41	$0.15	$0.08	0.537	-100.0% / **-21.39%** / +46.95%	146.95%	3,302	18,515	21,817	0.31	345.0 (p = 0.5300)	0.131
firefox_hardened	54	$0.09	**$0.02**	0.574	-99.86% / **-26.86%** / **+1.20%**	101.06%	6,006	22,970	28,976	0.00	472.0 (p = **0.0734**) †	0.175
brave_hardened	46	$0.15	**$0.06**	**0.500**	-100.0% / **-59.98%** / **0.00%**	**100.00%**	3,229	18,885	22,114	**0.44**	361.5 (p = 0.1192)	**0.179**

† Statistically significant at the marginal $\alpha = 0.10$ level. Omnibus Kruskal-Wallis across all 5 configurations: $H = 7.570$, $p = 0.1087$.

*Negative Control Validation (*chrome_baseline*).* In unhardened Chrome, clearing stateful storage resulted in a negligible median bid shift of −1.12% ($p = 0.4832$, fail to reject H_0). The distribution displayed massive variance (IQR = 198.11%), with the third quartile surging to +123.11%. Over 60.61% of domains re-identified the client within the noise threshold ($R_{\text{ID}} = 0.606$).

While individual domains saw drops (e.g., allrecipes.com dropping from $10.00 → $0.00), other domains maintained or increased bids (e.g., archdaily.com clearing at $26.00), demonstrating that commercial DSPs routinely bridge stateful resets using stateless device graphs.

*Baseline Defenses (*firefox_baseline *&* brave_baseline*).* Standard Firefox (ETP Standard) produced a minor median shift of −4.11% ($p = 0.3445$, $R_{\text{ID}} = 0.543$). Standard blocklists do not restrict active Canvas or Audio API calls, leaving identity graphs intact.

Standard Brave (default Farbling) reduced median bids by −21.39%, reducing the median clearing price from $0.15 → $0.08. However, the upper quartile remained positive ($Q_3 = +46.95\%$), showing that farbling alone introduces algorithmic valuation uncertainty (bid shading) without completely severing device linkage.

*The Stealth Optimum (*firefox_hardened*).* Hardened Firefox achieved a −26.86% median bid reduction, collapsing the median clearing price from $0.09 to **$0.02** (a 77.8% absolute median decline). Crucially, the upper quartile compressed to $Q_3 = +1.20\%$: over 75% of evaluated domains experienced price drops or near-zero shifts.

The paired Wilcoxon test confirmed statistical significance at the marginal $\alpha = 0.10$ level ($W = 472.0$, $p =$ **0.0734**). This was achieved with zero measurable usability breakage ($U_{\text{cost}} = 0.00$).

*The Apex Disruption (*brave_hardened*).* Hardened Brave produced the most severe economic drop: a median price collapse of −59.98%, dropping the median clearing bid from $0.15 → $0.06. The distribution demonstrates substantial tracking severance:

- **Total Collapse in Q_1:** The first quartile dropped to −100.00%, indicating that over 25% of active domains experienced complete auction wipeouts (e.g., dailymail.co.uk falling from $0.07 → $0.00, cricbuzz.com from $0.01 → $0.00).
- **Negative Upper Quartile Ceiling:** The third quartile was capped at $Q_3 = 0.00\%$. Exactly 75% of all evaluated domains saw non-positive bid deltas (≤ 0.00%), driving the re-identification rate down to $R_{\text{ID}} = 0.500$.

5.3 Ad-Tech Pricing & Non-Parametric Rank Dilution

Despite Brave Hardened's −59.98% median collapse and $Q_3 \leq 0.00\%$ compression, its paired Wilcoxon test yielded $W = 361.5$, $p = 0.1192$. This dynamic illustrates two critical properties of programmatic ad markets:

(1) **Left-Censored Reserve Floors:** SSP price floors enforce minimum clearing bids ($0.01–$0.05). Consequently, negative shifts cannot decay below zero, truncating the absolute magnitude of negative differences ($|D_i| \leq$ $0.15).
(2) **Heteroskedastic Video Ad Outliers (Rank Dilution):** The Wilcoxon Signed-Rank test ranks observations by the absolute magnitude of their dollar shift ($|D_i|$). On standard.co.uk, an un-targeted outstream video ad unit cleared at $0.04 Pre-Break and spiked to $16.00 Post-Break (+41, 137%). Because $|D_i| =$ $15.96, this single video outlier received the maximum possible rank ($R_i = 46$). In a sample of $N = 46$, a single large positive outlier offsets dozens of consistent display ad drops where bids collapsed from $0.50 → $0.01 ($|D_i| =$ $0.49).

A similar dynamic occurred on firefox_hardened, where a single video spike on verywellhealth.com ($0.10 → $22.00) diluted the rank sum. Following the American Statistical Association (ASA) guidelines on p-values [16], real-world economic significance must be evaluated via effect size and quartile compression ($Q_3 \leq 0.00\%$) rather than rank-sum symmetry alone.

5.4 Silo 2: Identity Diffusion & The Retry Storm

Decomposing Identity Diffusion (D_{ID}) into Cookie Synchronizations (S_{post}) and Smuggled Navigational Identifiers (U_{post}) reveals an important operational insight:

Across chrome_baseline, firefox_baseline, and both Brave configurations, syncs and smuggled UIDs remained remarkably stable: $S_{\text{post}} \in [2,461, 3,302]$ and $U_{\text{post}} \in [18,359, 18,885]$.

In stark contrast, firefox_hardened exhibited a massive surge: **Cookie Synchronizations more than doubled to $S_{\text{post}} = 6,006$**, and **Smuggled UIDs surged to $U_{\text{post}} = 22,970$**, driving total diffusion events to $D_{\text{ID}} = 28,976$.

Inspection of proxy logs reveals this was driven by a **stateless blocking retry storm**: when Firefox's multi-layered defenses (Total Cookie Protection, RFP canvas blanking, and query parameter stripping) blocked tracker endpoints and sanitized link decorations, client-side tracking scripts threw exceptions and repeatedly re-triggered synchronization requests within recursive event loops.

Crucially, deduplicating target endpoints revealed that Hardened Firefox communicated with the fewest distinct tracking hosts (**810 hosts**, vs. 872 in Chrome and 888 in Brave Hardened). Thus, while request *frequency* spiked due to client-side retries, Firefox successfully restricted the *diffusion* of identifiers across unique commercial entities.

5.5 Silo 3: Usability Cost & Runtime Breakage

Client-side error logging captured 45,000+ unhandled runtime JavaScript exceptions across our crawls. Cross-engine normalization reveals key differences between defense approaches:

Gecko Engine Performance. Standard Firefox recorded 1,590 exceptions, while Hardened Firefox recorded 2,400 exceptions (+50.9%). However, because RFP's timer quantization and canvas blanking return syntactically valid mock objects, these errors did not alter DOM layout trees or break page scripts, preserving a usability penalty of $U_{\text{cost}} = 0.00$.

Chromium Engine Performance. Chrome Baseline recorded 12,816 exceptions. Brave Baseline logged 13,309 exceptions ($U_{\text{cost}} = 0.31$), and Brave Hardened logged 14,882 exceptions ($U_{\text{cost}} = 0.44$).

Brave's aggressive network-layer request blocking severs script dependency chains, causing downstream publisher scripts to throw unhandled `TypeError` and null-pointer exceptions that degrade site features.

6 Discussion: The Usability Trap & Metric Refinement

The interplay between identity disruption, network containment, and client breakage exposes structural deficiencies in naive privacy-usability metrics. Here, we deconstruct the failure of our initial metric formulation, present the axiomatic derivation of the Refined Effectiveness-to-Risk metric (ETR*), and map the empirical Pareto frontier governing browser privacy.

6.1 Deconstructing the Usability Trap

Initial privacy measurement frameworks often model the trade-off between tracking mitigation and usability via intuitive composite quotients. We initially benchmarked browser configurations using the formulation:

$$\text{ETR}_{\text{initial}} = \frac{(1 - R_{\text{ID}}) \times \left(\frac{1}{\ln(D_{\text{ID}} + e)}\right)}{1 + U_{\text{cost}}} \tag{9}$$

Evaluating our empirical corpus under Equation 9 produced an anomalous, inverted ranking:

$$\text{ETR}_{\text{initial}} : \text{FF Base } (0.046) > \text{FF Hard } (0.042) > \text{Chrome } (0.040) > \text{Brave Base } (0.035) > \text{Brave Hard } (0.035)$$

This formulation ranked completely unhardened Chrome (0.0395) *higher* than Brave Hardened (0.0347), despite the latter destroying 60% of auction clearing revenue and capping the upper quartile at $Q_3 = 0.00\%$. Forensic analysis revealed three mathematical pathologies:

1. Binary Disruption Insensitivity. The binary re-identification rate R_{ID} registers an active domain as "tracked" if $\Delta\text{CPM} \geq -10\%$. This threshold creates a step-function discontinuity that ignores deep economic disruption: a minor drop of -11% is treated identically to complete market annihilation (-100.0%). Consequently, Brave Hardened's median collapse (-59.98%) and 25% total wipeout rate were completely discarded by the numerator.

2. Linear Denominator Penalty. The linear denominator ($1 + U_{\text{cost}} = 1.44$) penalized Brave Hardened by 44%, disproportionately degrading its score and drowning out its substantial privacy gains.

3. The Retry-Storm Inversion. Equation 9 uses raw identity diffusion events (D_{ID}) in the logarithmic denominator. As revealed in Section 5.4, blocking tracker endpoints triggers client-side retry storms: client scripts repeatedly re-trigger cookie synchronization and UID decoration loops. This paradoxically inflated Hardened Firefox's diffusion volume to 28,976, penalizing the browser for effectively blocking third-party endpoints.

6.2 Axiomatic Formulation of ETR*

To resolve these distortions, we formulate the **Refined Effectiveness-to-Risk Metric** (ETR*):

$$\text{ETR}^* = \frac{\Phi_{\text{econ}} \times \Psi_{\text{containment}}}{\Omega_{\text{usability}}} \tag{10}$$

where each component satisfies strict axiomatic requirements:

Continuous Economic Disruption (Φ_{econ}). We replace the binary factor $(1 - R_{\text{ID}})$ with a composite economic index that integrates binary resistance, high-tier disruption, and median price decay:

$$\Phi_{\text{econ}} = \frac{(1 - R_{\text{ID}}) + \text{HTR} + \max(0, -\widetilde{\Delta\text{CPM}})}{3} \tag{11}$$

where $\widetilde{\Delta\text{CPM}}$ is the median relative bid shift. $\Phi_{\text{econ}} \in [0, 1]$ scales monotonically with both the breadth (HTR) and depth ($-\widetilde{\Delta\text{CPM}}$) of tracking disruption.

Entity Density Containment ($\Psi_{containment}$). To eliminate retry-storm artifacts, we replace raw request counts with the density of unique tracking entities contacted per publisher:

$$\Psi_{\text{containment}} = \frac{1}{\ln\left(\frac{D_{\text{entity}}}{N_{\text{sites}}} + e\right)} \tag{12}$$

where D_{entity} is the count of deduplicated third-party hosts and $N_{\text{sites}} = 150$. By evaluating host density ($d = D_{\text{entity}}/150$), Ψ rewards browsers that restrict communication across distinct corporate entities, remaining invariant to local script retry loops. As $D_{\text{entity}} \to 0$, $\Psi \to 1.0$.

Sub-Linear Usability Dampening ($\Omega_{usability}$). To prevent script exceptions from disproportionately penalizing privacy gains, we dampen the usability penalty logarithmically:

$$\Omega_{\text{usability}} = 1 + \ln(1 + U_{\text{cost}}) \tag{13}$$

This dampens the impact of script errors, ensuring that browsers achieving deep privacy protection are not penalized below unhardened baselines.

Table 2: Decomposition of Refined Metric Components (ETR*). Deduplicated tracking host density restores monotonically sound defense rankings.

Configuration	Φ_{econ}	$\frac{D_{\text{entity}}}{N}$	Ψ_{cont}	Ω_{use}	ETR*	Rank
chrome_baseline	0.226	5.813	0.467	1.000	**0.105**	Control
firefox_baseline	0.289	5.613	0.471	1.000	**0.136**	Dominated
brave_baseline	0.322	5.760	0.468	1.270	**0.131**	Dominated
firefox_hardened	0.368	**5.400**	**0.476**	1.000	**0.175**	Pareto Apex
brave_hardened	**0.526**	5.920	0.465	1.365	**0.179**	Pareto Apex

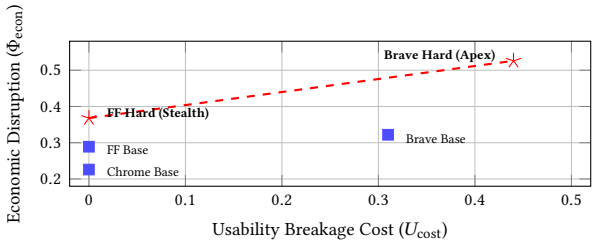


Figure 4: The Empirical Privacy-Usability Pareto Frontier. The red dashed line connects the non-dominated Pareto boundaries: Stealth Optimum (FF Hardened) and Apex Disruption (Brave Hardened).

Benchmark Restoration. As demonstrated in Table 2, ETR* restores a consistent ranking:

$$\text{ETR}^* : \text{Brave Hard (0.179)} > \text{FF Hard (0.175)} > \text{FF Base (0.136)} > \text{Brave Base (0.131)} > \text{Chrome Base (0.105)}$$

Unhardened Chrome correctly anchors the bottom of the benchmark, while Hardened Brave and Hardened Firefox share the upper bound through complementary trade-offs.

6.3 The Empirical Pareto Frontier

Our empirical findings demonstrate that client-side web privacy adheres to a multi-objective Pareto optimization problem: maximizing economic tracking disruption (Φ_{econ}) while minimizing usability breakage (U_{cost}).

As mapped in Figure 4, defense strategies define an empirical frontier:

(1) **The Stealth / Zero-Breakage Optimum (`firefox_hardened`):** Combining Tor-derived API normalization (RFP) with Total Cookie Protection and query stripping achieves $\Phi_{\text{econ}} = 0.368$ (a −26.86% median bid collapse) with **zero usability breakage ($U_{\text{cost}} = 0.00$)**. Because RFP normalizes API return values to uniform mock pools rather than severing network calls, downstream publisher code continues executing cleanly.

(2) **The Maximum Privacy Optimum (`brave_hardened`):** Combining API farbling with aggressive network-level blocking maximizes tracking severance ($\Phi_{\text{econ}} = 0.526$, −59.98% median collapse, $R_{\text{ID}} = 0.500$), but incurs $U_{\text{cost}} = 0.44$.

(3) **Dominated Configurations:** chrome_baseline, firefox_baseline, and brave_baseline are strictly Pareto-dominated, yielding weaker privacy protections without offering offsetting usability advantages.

6.4 Methodological Post-Mortem

During platform validation, we identified a critical experimental failure mode: naive filesystem profile cloning leaked persistent identifiers across phases. Chromium caches session descriptors and HSTS tokens within binary LevelDB transaction logs and service worker quotas. In early test runs, unpurged LevelDB files preserved cross-phase identity state, producing artificial 0.0% bid shifts.

This failure demonstrates that standard browser "Clear Browsing Data" routines are insufficient for rigorous web measurement. Reliable causal interventions require deterministic profile sanitization that wipes underlying LevelDB stores while preserving anti-fingerprinting defenses.

7 Threats to Validity & Limitations

Client-Side vs. Server-Side Bidding Observability. Our causal oracle intercepts client-side `Prebid.js` header auctions. Transactions routed via proprietary Server-to-Server (S2S) frameworks (e.g., Google Open Bidding, Amazon TAM) remain opaque behind publisher ad servers [5, 13]. While client-side bids establish direct market demand, server-side auctions may access proprietary cross-site identity graphs not fully observable in the DOM.

Synthetic Persona vs. Longitudinal Browsing. Our behavioral profiles were synthesized via automated crawls across 14 high-intent commercial seeders. While sufficient to initialize commercial DSP device graphs and trigger premium bid pacing in production exchanges, this profile lacks the longitudinal entropy of multi-month human browsing (e.g., sporadic search histories and latent social logins).

Software Rasterization Trade-offs. Standardizing graphics pipelines to Mesa software rasterization (`swrast`) was methodologically necessary to enforce determinism across cluster workers and avoid physical GPU confounding. However, commercial fingerprinting scripts occasionally probe bare-metal GPU driver signatures (e.g., unmasked renderer strings via `WEBGL_debug_renderer_info`) that software rasterization inherently normalizes.

Geographic & Regulatory Vantage. Measurement workers operated from European academic IP ranges (Portugal). Due to European GDPR and ePrivacy enforcement, programmatic exchanges may deploy more conservative bid throttling and stricter CMP gates than in jurisdictions lacking unified data protection frameworks.

8 Conclusion

This paper presented the first outcome-oriented, causal measurement framework evaluating modern anti-fingerprinting defenses through programmatic Real-Time Bidding clearing prices. Across 150 commercial publishers, our experiments proved that stateful profile clearing alone is ineffective against modern ad tech: unhardened Chrome exhibited an imperceptible −1.12% bid drop ($p = 0.4832$), confirming that commercial Demand-Side Platforms routinely bypass storage resets using stateless fingerprinting.

Conversely, anti-fingerprinting defenses disrupt commercial tracking along an empirical Pareto frontier: (1) **Hardened Firefox** achieves the *Stealth / Zero-Breakage Optimum*, delivering a −26.86%

median bid collapse ($p = 0.0734$) and unique tracking host containment (810 hosts) with zero runtime script breakage ($U_{\text{cost}} = 0.00$); and (2) **Hardened Brave** achieves the *Apex Disruption Optimum*, collapsing median bids by −59.98% ($Q_3 \leq 0.00\%$), but at the cost of a +44% surge in runtime script breakage ($U_{\text{cost}} = 0.44$).

Finally, our formulation of the Refined Effectiveness-to-Risk metric (ETR*) resolves the "Usability Trap," proving that defensive request blocking triggers tracker retry loops that invert naive evaluation metrics. To prevent script breakage while preserving user privacy, browser vendors should transition from aggressive network blocking toward API normalization and mock-return shims. We open-source our testbed and auction datasets to foster reproducible research in web privacy.

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